

farmers independent of the grid supply, and also enable them to sell surplus solar power generated to dis-coms and get extra income

- Solarisation of tubewells and lift irrigation projects of the government sector

Floating solar, an emerging trend

Floating solar, an emerging and innovative application of solar PV technology, was conceived to overcome land scarcity issues. It can be deployed on various types of water bodies including industrial water ponds, irrigation or drinking water reservoirs, quarry lakes, aquaculture ponds, canals and dams. Most of the deployment is on fresh water until the challenges associated with marine environment (high turbulence and salt mist corrosion) are overcome. Key drivers making floating solar attractive are:

- It solves land scarcity issues and negates the need for expensive and time-consuming land acquisition process.
- It can use existing grid infrastructure. For example, a floating solar project located near a consumption centre or on a reservoir of a hydro-electric dam can work with the existing transmission infrastructure.
- It provides efficient electricity generation, as cooling effect from water surface on the panels results in three to five per cent more power output as compared to ground-mounted solar modules.

China is the leading international market, followed by Japan and South



Floating solar power in India (Credit: www.vikramsolar.com)

Impact of Union Budget 2019-20 on the solar sector

- Total budgetary allocation for the renewable sector in 2019-20 is ₹ 52.5 billion. This is only two per cent higher than the 2018-19 allocation.
- Almost sixty per cent (₹ 30 billion) of the total budget has been designated for the solar sector.
- ₹ 24.8 billion will be for 7500MW grid-connected solar projects, and the rest will be for off-grid projects.
- The government is focusing on boost solar cell manufacturing.

Korea. This concept is still in the nascent stages of development in India. According to a PHOTON report, after the recent commissioning of a 2MW project at Visakhapatnam, cumulative capacity of floating solar reached 2.7MW in the country. MNRE intends to add 10GW of floating solar as part of its 227GW renewable energy target for 2022.

Moving forward

Sustainable growth of the solar energy industry depends on successful management of PV waste. PV waste

recycling is still at a nascent stage globally, both in terms of technical standards and physical infrastructure. Use of potentially-hazardous materials in manufacturing, and lack of commercially-viable module recycling technologies, demand a strong regulatory approach to this problem.

The European Union (EU) has taken the lead with a comprehensive plan for allocating liability, setting recovery and recycling targets, treatment requirements and consumer awareness.

India is still not well-equipped to handle PV waste as there are no policy guidelines available for the same. Even e-waste management rules in the country do not mention anything about solar PV waste. Thus, lack of a policy framework coupled with unavailability of basic PV recycling technologies in the country have worsened the situation.

As per Central Pollution Control Board's (CPCB's) estimate, less than four per cent of estimated e-waste is recycled in the organised sector. Thus, it is imperative for India to create a robust regulatory framework that clearly mentions the responsibility and specify standards for PV waste management. **EFY**

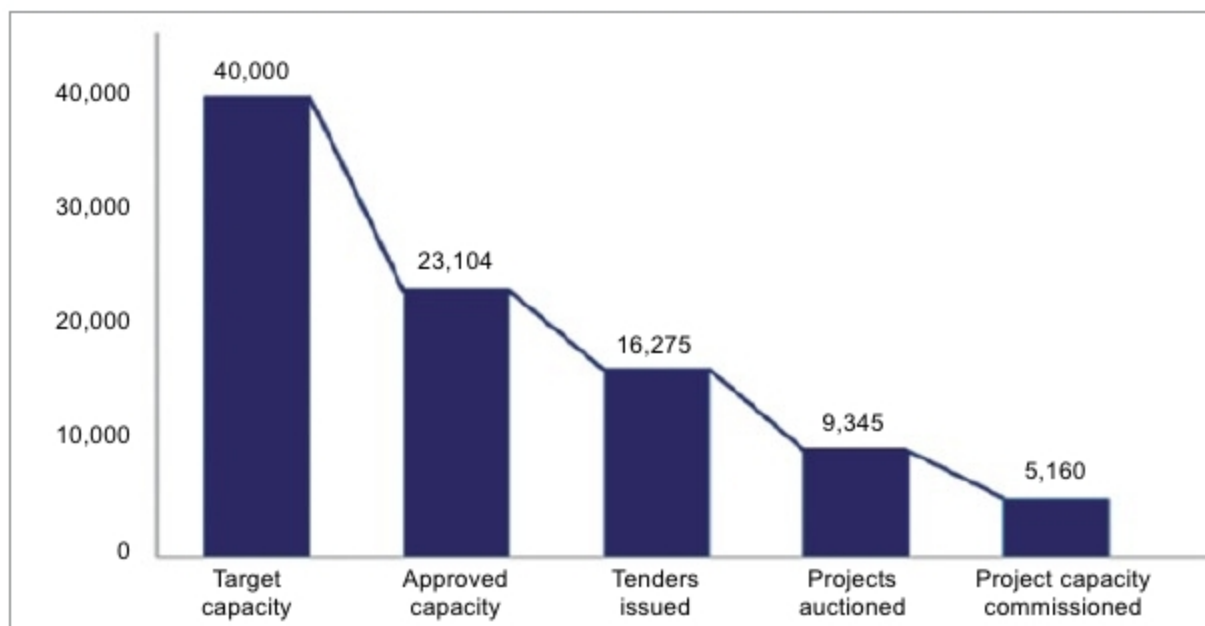


Fig. 4: Capacity-wise (MW) solar park development status in India till the end of 2018-19 (Source: Bridge To India)